```
#include <iostream>
#include <iomanip>
#include <limits>
#include <string>
using namespace std;

constexpr long long MIN_BALANCE_PAISE = 5000;

bool readNumber(const string& prompt, long long& value) {
  while (true) {
    cout << prompt;

    if (cin >> value) {
      return true;
    }

    if (cin.eof()) {
      return false;
    }

    cout << "Invalid input! Please enter a number.\n";

    cin.clear();
    cin.ignore(numeric_limits<streamsize>::max(), '\n');
  }
}

void printMoney(long long paise) {
```

```
    cout << "Rs. " << fixed << setprecision(2)
         << paise / 100.0;
}

int main() {
    string cardId;
    long long balance, fare, journeyCount = 0;

    cout << "NAMMA METRO SMART-CARD SYSTEM\n";

    cout << "Enter Card ID: ";
    if (!(cin >> cardId)) {
        cout << "Invalid card ID input.\n";
        return 1;
    }

    if (!readNumber("Enter initial balance in paise: ", balance)) {
        cout << "Input ended unexpectedly.\n";
        return 1;
    }

    if (balance < 0) {
        cout << "Balance cannot be negative.\n";
        return 1;
    }

    while (true) {
        if (!readNumber(
```

```
      "Enter fare in paise (0 to finish): ", fare)) {
   cout << "\nInput ended unexpectedly.\n";
   break;
}

if (fare == 0) {
   break;
}

if (fare < 0) {
   cout << "Fare cannot be negative. Try again.\n";
   continue;
}
if (balance < fare ||
   balance - fare < MIN_BALANCE_PAISE) {
   cout << "Journey denied: minimum balance "
        << "of Rs. 50 must remain.\n";
   continue;
}

balance -= fare;
journeyCount++;

cout << "Journey accepted. Fare: ";
printMoney(fare);
cout << '\n';

cout << "Remaining balance: ";
```

```
        printMoney(balance);
        cout << "\n";
    }

    cout << "\n========== JOURNEY REPORT ==========\n";
    cout << "Card ID: " << cardId << '\n';
    cout << "Successful journeys: " << journeyCount << '\n';
    cout << "Final balance: ";
    printMoney(balance);
    cout << "\n====================================\n";

    return 0;
}
```